

TriVoKo

Phase 8 - Returns & refunds

Time	Start date	Finish date	Status
1.5 weeks			Not started

Goal: Standout #2: a customer returns one item and gets back exactly the right money through a Stripe partial refund, while stock and the seller ledger stay correct.

Real-life example: You return one shirt from a bill of three. The shop checks it, takes it back to the shelf, and returns only that shirt's money - minus its part of the discount you used.

Step checklist

★ = you type the key logic (Claude explains first, then reviews). "You" = only you can do it.

#	Step	Who	Done
1	Flyway: return_requests, refunds	Claude	☐
2	★ You type the RETURN STATE MACHINE: ReturnStatus enum with allowedNext() and the service method that moves a return	★ Madhavan	☐
3	★ You write tests for allowed and illegal moves	★ Madhavan	☐
4	Request a return: DELIVERED items only, within 7 days, once per item, reason + optional photo	Claude	☐
5	Seller approve / reject with reason; auto-approve job after 2 days	Claude	☐
6	Escalate once; admin makes the final decision	Claude	☐
7	Picked up → Stripe partial refund of (price × qty – coupon share), one idempotency key per return	Claude	☐
8	Refund done → stock back, minus line in the seller ledger, email + bell	Claude	☐
9	Frontend: return form + timeline (customer), returns page (seller), disputes page (admin); return rate in admin reports	Claude	☐
10	Tests: refund maths, double click = one refund	Claude	☐

Output of this phase

- Complete returns flow for 3 roles
- Stripe partial refunds
- Your ReturnStatus state machine + tests

Done when (check before moving on)

Check	Passed
Return one item of a 2-item order → the exact partial refund appears in the Stripe test dashboard	☐

Check	Passed
Ledger and stock match	☐

Explain it back (in your own words)

Question	Done
Draw the return states and say who can make each move	☐
How is the refund amount calculated?	☐
Why can a refund never happen twice?	☐

Finish routine

Check	Done
All tests pass and CI is green	☐
Committed and pushed	☐
Done-when checks shown working in the browser	☐
Session summary + SESSION_LOG written	☐

Skills you practise: State machines, Stripe refunds, idempotency, scheduled jobs

Where your work goes: `backend\...\returns`, `...\payment`, return pages in `frontend`

My notes
